

Description

While the patient is prepared for transport within the hospital for ECMO, the clinical staffers complete an admission record. The Reason for admission is "The Abdominal scan shows the newborn has Pericardial and diaphragmatic defect syndrome.". The Admit Diagnosis is recorded as, "Congenital diaphragmatic hernia (CDH)".

NIST Hospital reports syndromic surveillance data to the City Health Department (NIST PH SS FAC 1). At 8:35 AM on February 1, 2025, the hospitals electronic health record module for syndromic surveillance data assembles and transmits an Admission message about this encounter. The "full, legal patient name" and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions.

Comments

Dates and times are provided in this test case to illustrate the sequence of clinical and messaging events. Since the exact dates and times are not reproducible when modeling the Test Case with an HIT Module, only date and time format will be validated within tester submitted test data.

Pre-condition

A03-Discharge message is sent before A01-Admission message.

Post-Condition

No Post-Condition

Test Objectives

This test step examines a Health IT Module's ability to create ADT^A01 Admission message within the HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements for syndromic surveillance.

- Setting is Emergency Department with Inpatient Admission.
- Chief complaint is captured as narrative in PV2-3 Admit Reason.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:** Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- **Patient Name is Identifiable in the submissions:** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). The patient name presented in an identified manner is required for this Test Step.
- Multiple Races.
- Pediatric case where age is measured in months.
- Addition of Admitting Diagnoses.
- Discharge Disposition Home.
- Insurance is unknown

Evaluation Criteria

Message Validation Report

Diagnosis code (DG1-3) validation is validated as follows:

The table below summarizes the format representations that are accepted for ONC Health IT Module Certification. Rows 1 and 2 show the formats as required by the implementation guide and standard terminology rules and best practices. The last 3 columns show acceptable alternative formats. For this test case one of the following values shall be present in the message:

DG1(1)-3

Code	Concept	Code System	DG1-3 Message Snippet	NIST
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Required Formats based on Recognized Terminology Standards Rules

P2401	Meconium aspiration with respiratory symptoms	I10C	P2401^Type III occipital condyle fracture, initial encounter for closed fracture^I10C	Yes
206292002	Meconium aspiration syndrome (disorder)	SCT	206292002^Meconium aspiration syndrome (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
P24.01	Type III occipital condyle fracture, initial encounter for closed fracture	I10C	P24.01^Type III occipital condyle fracture, initial encounter for closed fracture^I10C	Yes
P2401	Type III occipital condyle fracture, initial encounter for closed fracture	I10	P2401^Type III occipital condyle fracture, initial encounter for closed fracture^I10	No
P24.01	Type III occipital condyle fracture, initial encounter for closed fracture	I10	P24.01^Type III occipital condyle fracture, initial encounter for closed fracture^I10	No
276533002	Neonatal aspiration syndrome (disorder)	SCT	276533002^Neonatal aspiration syndrome (disorder)^SCT	Yes
DG1(2)-3				
Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
Q790	Congenital diaphragmatic hernia	I10C	Q790^Congenital diaphragmatic hernia^I10C	Yes
17190001	Congenital diaphragmatic hernia (disorder)	SCT	17190001^Congenital diaphragmatic hernia (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
Q79.0	Congenital diaphragmatic hernia	I10C	Q79.0^Congenital diaphragmatic hernia^I10C	Yes
Q790	Congenital diaphragmatic hernia	I10	Q790^Congenital diaphragmatic hernia^I10	No
Q79.0	Congenital diaphragmatic hernia	I10	Q79.0^Congenital diaphragmatic hernia^I10	No
88386004	Congenital anomaly of diaphragm (disorder)	SCT	88386004^Congenital anomaly of diaphragm (disorder)^SCT	Yes
DG1(3)-3				
Code	Concept	Code System	DG1-3 Message Snippet	b
Required Formats based on Recognized Terminology Standards Rules				
Q791	Other congenital malformations of diaphragm	I10C	Q791^ Other congenital malformations of diaphragm^I10C	Yes

724068001	724068001 Pericardial and diaphragmatic defect syndrome (disorder)	SCT	724068001^Pericardial and diaphragmatic defect syndrome (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
Q79.1	Other congenital malformations of diaphragm	I10C	Q79.1^Other congenital malformations of diaphragm^I10C	Yes
Q791	Other congenital malformations of diaphragm	I10	Q791^Other congenital malformations of diaphragm^I10	No
Q79.1	Other congenital malformations of diaphragm	I10	Q79.1^Other congenital malformations of diaphragm^I10	No
39839004	Congenital anomaly of diaphragm (disorder)	SCT	39839004^Congenital anomaly of diaphragm (disorder)^SCT	Yes

The NIST column indicates what the NIST Tool Validation will accept in context-based testing (used for ONC Certification). A 'Yes' in the NIST column signifies that the tool will accept the format and not throw an error. A 'No' in the NIST column signifies that the tool will throw an error (specifically for DG1-3.3). However, based on the recommendation from the CDC SMEs and approval from ASTP/ONC the format is acceptable. The NIST test cases documents the instances where the test proctor can grant an exception when the tool issues and error.

Notes

PV1-2 may be populated with either "E" or "I".

An OBX segment in the A01 message provides the originating visit type information through Facility/Visit Type. The admit described in the ED_Visit_Admit Test Scenario originated in the emergency department; therefore, "261QE0002X^Emergency Care^HCPTNUCC" is in the OBX.5 field in the Example A01 Message. Per ONC, the OBX.5 field may be populated with "1021-5^Inpatient practice setting^HSLOC" for certification testing. The Tool performs automated validation to check for either of those values.

Codes are provided in the test data for coding the diagnoses and procedures. The Tool is designed to accept those codes without generating an error. If the HIT developer uses a different but equivalent coded value within the same code system than the ones provided/accepted, the Tester may ignore errors generated by the Test Tool related to incorrect code/name of coding system when the coded data used in the message are determined via visual inspection to be valid data.

Messages in this test case will be used in a Manual Test Step under functional requirement testing.